



CLIPGEAR - Infinite Modular Snap-Together Gear System



Abílio Costa

[VIEW IN BROWSER](#)

updated 22. 8. 2026 | published 22. 8. 2026

Summary

An infinitely expandable, snap-together gear system for endless mechanical creations.

[Learning](#) > [Engineering](#)

Tags: [modular](#) [fidget](#) [gear](#) [gears](#) [mechanical](#)
[educational](#) [engineering](#) [stem](#) [snapfit](#) [clipgear](#)

CLIPGEAR is an infinitely expandable, modular gear system. Snap the bases together, clip the gears on, and build your own mechanical creations - or simply use it as a **satisfying fidget**!

The system is designed to be **simple, lightweight, and endlessly expandable**. It supports **1:2 and 2:1 gear ratios**, allowing you to trade speed for torque, or torque for speed. It also includes a variety of gears and special parts, including gears with a **wheel** and a **fan**, with more possibilities to come.

Gear Sizes & Files

Gears are available in **16, 24, and 32 teeth**. Files are named Gear X-Y, where:

- **X** = bottom gear
- **Y** = top gear

For example, Gear 16-32 contains a 16-tooth bottom gear and a 32-tooth top gear.

Printing

- **No supports** required
- **0.20 mm layer height** recommended
- **PLA or PETG** both work well. **PETG** on the base makes the connections easier
- **Low infill** required

Alternative bases

Fixed-size bases and Multiboard-compatible bases are also available:

Updates

2026-08-22

- Added a variation of the base with a tighter gear pin, to reduce wobbly.
- Added a new two-part **crank**! No supports needed!

Model files

 **Bases** 2 files

 **clipgear-base-tighter-gear-pin.stl**
☐ Recommended. If too tight on the gears, use the looser version



clipgear-base-looser-gear-pin.stl

☐ Use this one if the tighter version is too tight on the gears



Gears

15 files



clipgear-gear-0-16.stl



clipgear-gear-24.stl



clipgear-gear-16.stl



clipgear-gear-16-16.stl



clipgear-gear-24-16.stl



clipgear-gear-32-32.stl



clipgear-gear-0-24.stl



clipgear-gear-32.stl



clipgear-gear-32-24.stl



clipgear-gear-24-24.stl



clipgear-gear-16-24.stl



clipgear-gear-0-32.stl



clipgear-gear-16-32.stl



clipgear-gear-24-32.stl



clipgear-gear-32-16.stl



Special gears

4 files



clipgear-crank-handle.stl



clipgear-special-16-24-crank-body.stl



clipgear-special-16-24-fan.stl



clipgear-special-24-24-wheel.stl

License

This work is licensed under a
[Creative Commons \(4.0 International License\)](https://creativecommons.org/licenses/by/4.0/)



Attribution

- ✗ | Sharing without ATTRIBUTION
- ✓ | Remix Culture allowed
- ✓ | Commercial Use
- ✓ | Free Cultural Works
- ✓ | Meets Open Definition